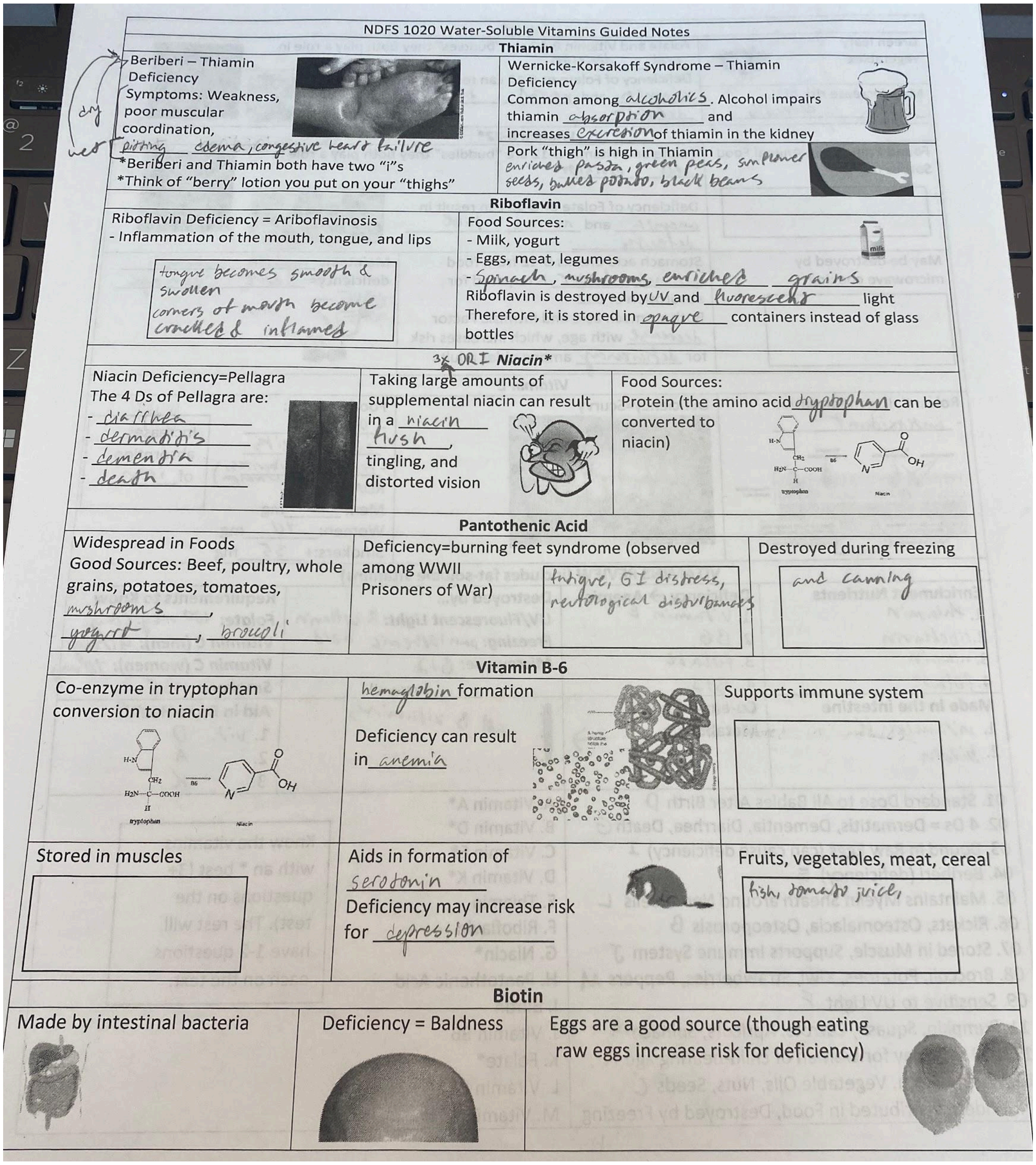
Know



NDFS 1020 Water-Soluble Vitamins Guided Notes

Thiamin

Beriberi – Thiamin Deficiency
Symptoms: Weakness, poor muscular coordination,
dry
wet
pitting edema, congestive heart failure
*Beriberi and Thiamin both have two "i"s
*Think of "berry" lotion you put on your "thighs"

Wernicke-Korsakoff Syndrome – Thiamin Deficiency
Common among alcoholics. Alcohol impairs thiamin absorption and increases excretion of thiamin in the kidney

Pork "thigh" is high in Thiamin
enriched pasta, green peas, sunflower seeds, baked potato, black beans

Riboflavin

Riboflavin Deficiency = Ariboflavinosis
- Inflammation of the mouth, tongue, and lips

tongue becomes smooth & swollen
corners of mouth become cracked & inflamed

Food Sources:
- Milk, yogurt
- Eggs, meat, legumes
- Spinach, mushrooms, enriched grains

Riboflavin is destroyed by UV and fluorescent light
Therefore, it is stored in opaque containers instead of glass bottles

3x OR I ***Niacin****

Niacin Deficiency=Pellagra
The 4 Ds of Pellagra are:
- diarrhea
- dermatitis
- dementia
- death

Taking large amounts of supplemental niacin can result in a niacin flush, tingling, and distorted vision

Food Sources:
Protein (the amino acid tryptophan can be converted to niacin)

Pantothenic Acid

Widespread in Foods
Good Sources: Beef, poultry, whole grains, potatoes, tomatoes, mushrooms, yogurt, broccoli

Deficiency=burning feet syndrome (observed among WWII Prisoners of War)
fatigue, GI distress, neurological disturbances

Destroyed during freezing
and canning

Vitamin B-6

Co-enzyme in tryptophan conversion to niacin

hemoglobin formation
Deficiency can result in anemia

Supports immune system

Stored in muscles

Aids in formation of serotonin
Deficiency may increase risk for depression

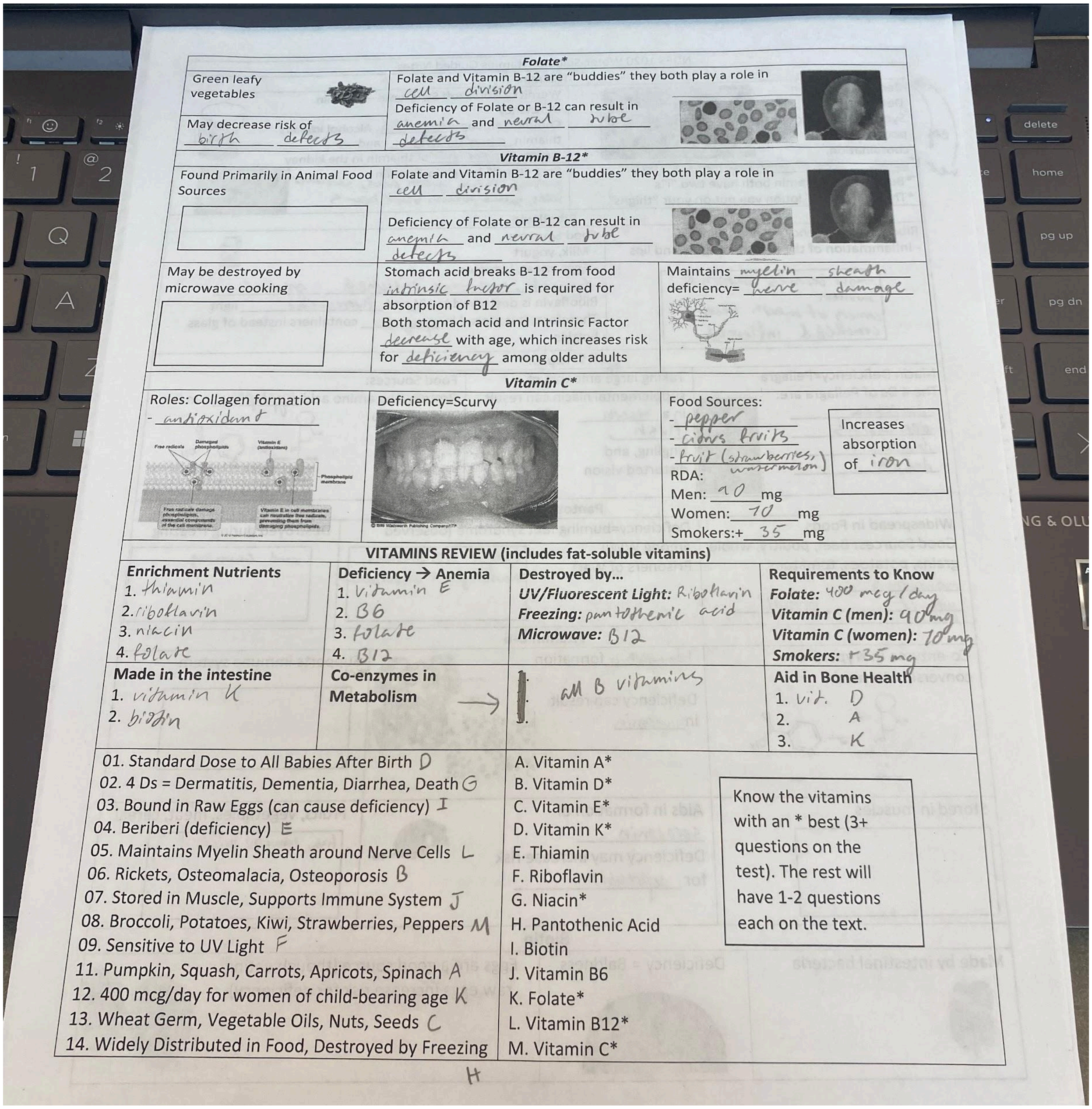
Fruits, vegetables, meat, cereal
fish, tomato juice,

Biotin

Made by intestinal bacteria

Deficiency = Baldness

Eggs are a good source (though eating raw eggs increase risk for deficiency)



Folate*

Green leafy vegetables

May decrease risk of birth defects

Folate and Vitamin B-12 are "buddies" they both play a role in cell division

Deficiency of Folate or B-12 can result in anemia and neural tube defects

Vitamin B-12*

Found Primarily in Animal Food Sources

Folate and Vitamin B-12 are "buddies" they both play a role in cell division

Deficiency of Folate or B-12 can result in anemia and neural tube defects

May be destroyed by microwave cooking

Stomach acid breaks B-12 from food
intrinsic factor is required for absorption of B12
Both stomach acid and Intrinsic Factor decrease with age, which increases risk for deficiency among older adults

Maintains myelin sheath
deficiency= nerve damage

Vitamin C*

Roles: Collagen formation
- antioxidant

Deficiency=Scurvy

Food Sources:
- pepper
- citrus fruits
- fruit (strawberries, watermelon)

RDA:
Men: 90 mg
Women: 70 mg
Smokers:+ 35 mg

Increases absorption of iron

VITAMINS REVIEW (includes fat-soluble vitamins)

Enrichment Nutrients	Deficiency → Anemia	Destroyed by...	Requirements to Know
1. thiamin 2. riboflavin 3. niacin 4. folate	1. Vitamin E 2. B6 3. folate 4. B12	*UV/Fluorescent Light:* Riboflavin *Freezing:* pantothenic acid *Microwave:* B12	*Folate:* 400 mcg/day *Vitamin C (men):* 90 mg *Vitamin C (women):* 70 mg *Smokers:* +35 mg
Made in the intestine 1. vitamin K 2. biotin	**Co-enzymes in Metabolism** →	all B vitamins	**Aid in Bone Health** 1. vit. D 2. A 3. K

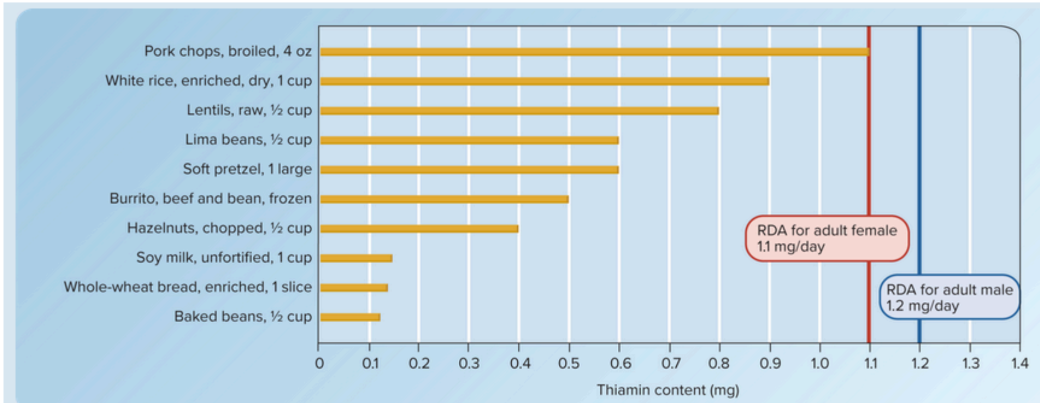
01. Standard Dose to All Babies After Birth D	A. Vitamin A*
02. 4 Ds = Dermatitis, Dementia, Diarrhea, Death G	B. Vitamin D*
03. Bound in Raw Eggs (can cause deficiency) I	C. Vitamin E*
04. Beriberi (deficiency) E	D. Vitamin K*
05. Maintains Myelin Sheath around Nerve Cells L	E. Thiamin
06. Rickets, Osteomalacia, Osteoporosis B	F. Riboflavin
07. Stored in Muscle, Supports Immune System J	G. Niacin*
08. Broccoli, Potatoes, Kiwi, Strawberries, Peppers M	H. Pantothenic Acid
09. Sensitive to UV Light F	I. Biotin
11. Pumpkin, Squash, Carrots, Apricots, Spinach A	J. Vitamin B6
12. 400 mcg/day for women of child-bearing age K	K. Folate*
13. Wheat Germ, Vegetable Oils, Nuts, Seeds C	L. Vitamin B12*
14. Widely Distributed in Food, Destroyed by Freezing H	M. Vitamin C*

Know the vitamins with an * best (3+ questions on the test). The rest will have 1-2 questions each on the text.

Thiamin

- Functions
    - Part of coenzyme thiamin pyrophosphate (TPP), used in breakdown of carbs for energy and for the metabolism of branched chain amino acids
        - Also used to synthesize neurotransmitters
- Food sources
    - Whole grain and enriched breads and cereals, pork, legumes, nuts, brewer's yeast

TABLE 10.2 Thiamin Content of Selected Foods

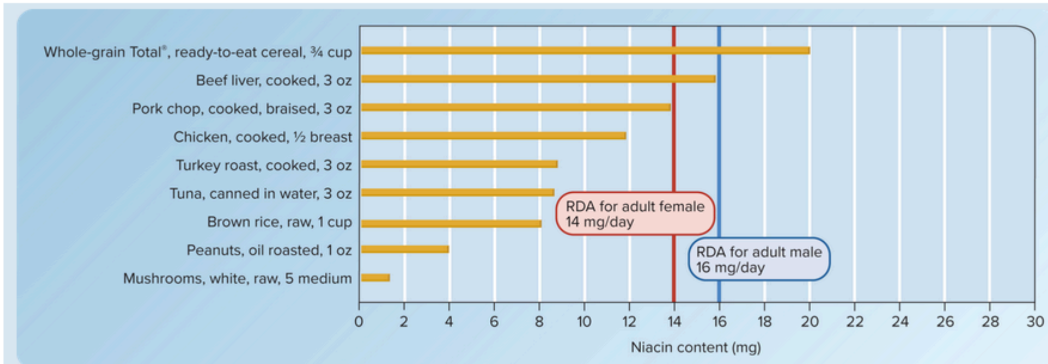
- 

    - Sensitive to overheating
- Dietary adequacy
    - RDA: 1.2 mg for males, 1.1 mg for females
    - Deficiency
        - Beriberi
        - Weakness
        - Poor muscular coordination
        - Negative impact on cardiovascular, digestive, and nervous systems in severe cases
        - Wernicke-korsakoff syndrome (in alcoholics)
            - Abnormal eye movements
            - Staggering gait
            - Distorted thought processes
            - Can lead to permanent disability or death if not treated
    - Toxicity
        - Rare, no UL

Niacin

- Functions
    - Used to synthesize coenzymes (NAD and NADP) involved in a lot of stuff, like the release of energy from macronutrients
- Food sources
    - Enriched cereals, beef liver, tuna, salmon, poultry, pork, mushrooms
    - Can be converted into it (tryptophan): eggs, cows milk
    - 60 mg tryptophan = 1 mg niacin

TABLE 10.4 Niacin Content of Selected Foods



- Dietary adequacy
    - RDA: 14-16 mg a day
    - Niacin equivalents:

Step 1:	Determine the grams of protein available for niacin synthesis (RDA for protein is 0.8g/kg body weight). The protein available for niacin synthesis is any amount consumed above that amount.
Step 2:	Assume that most sources of dietary protein contain approximately 1% tryptophan. Based on this assumption, divide the grams of excess daily protein intake by 100 to obtain the grams of tryptophan. Convert this amount to milligrams (multiply by 1000).
Step 3:	Because the body can convert 60 mg of tryptophan into 1 mg of niacin, divide the milligrams of tryptophan from step 2 by 60. This indicates the NEs available from tryptophan.
Step 4:	Add together the amount of food niacin and niacin available from tryptophan (step 3). This amount provides an estimate of dietary niacin.

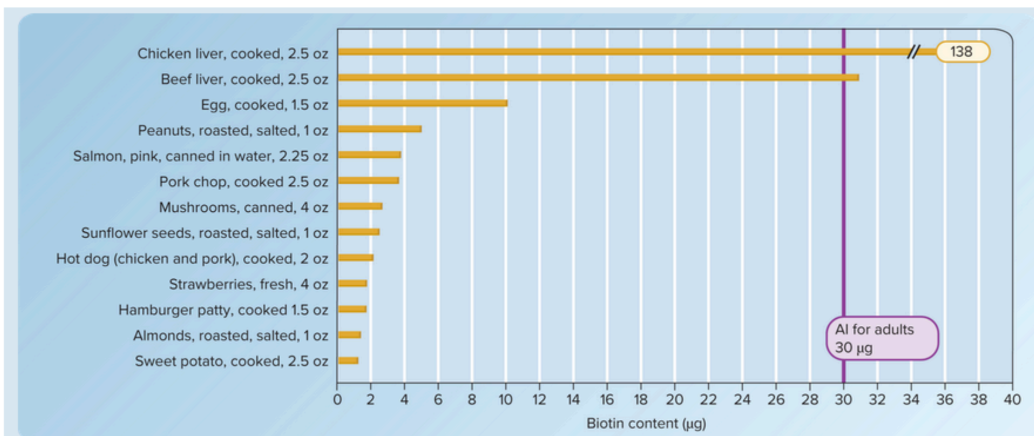
- Deficiency
    - At risk:
        - Alcoholics
        - Eating disorders
        - Disorders that disrupt tryptophan metabolism
    - Poor appetite
    - Weight loss
    - Weakness
    - Eventually worsens into pellagra (four Ds)
        - Dermatitis
        - Diarrhea
        - Dementia
        - Death
- Toxicity
    - UL = 35 mg
    - Can be achieved with megadoses, which are sometimes used for reducing LDL levels and increasing HDL levels.
        - May increase risk of stroke or heart attack in patients that cannot tolerate statins

        - Flushing of skin
        - Nausea, vomiting, GI tract upset
        - Liver damage

Biotin

- Functions
    - Chemical reactions to add CO_2 to other compounds
        - Synthesis of glucose and fatty acids
        - Breakdown of certain amino acids
    - Essential coenzyme for regenerating oxaloacetate in the citric acid cycle
- Food sources
    - Liver, eggs, peanuts, salmon, pork

      TABLE 10.6 Biotin Content of Selected Foods

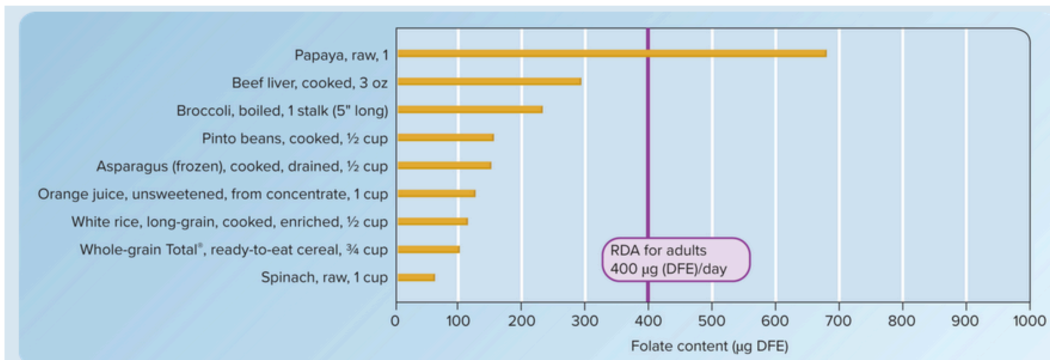
    -
    - NOT RAW EGG WHITES. Opposite
- Dietary adequacy
    - AI = 30 micrograms
    - Deficiency
        - Rare, intestinal bacteria can make some and is widely found in foods
        - Skin rash
        - Hair loss
        - Convulsions
        - Neurological disorders
        - Developmental delays in infants
    - Toxicity
        - No UL, nontoxic

Folate

- Functions
    - DNA synthesis
    - Amino acid metabolism

- Food sources
    - Leafy veggies, liver, legumes, asparagus, broccoli, orange juice, enriched grain products, fortified cereals

TABLE 10.8 Folate Content of Selected Foods



    -
    - Destroyed easily by heat, oxidation, and UV light

Calculating DFEs

The following equation is used to determine DFEs:

DFE = µg natural food folate + (1.7 × µg folic acid from fortified foods or dietary supplements)

If a person consumes 100 µg of food folate and 200 µg of folic acid from supplements or fortified foods, their DFE is

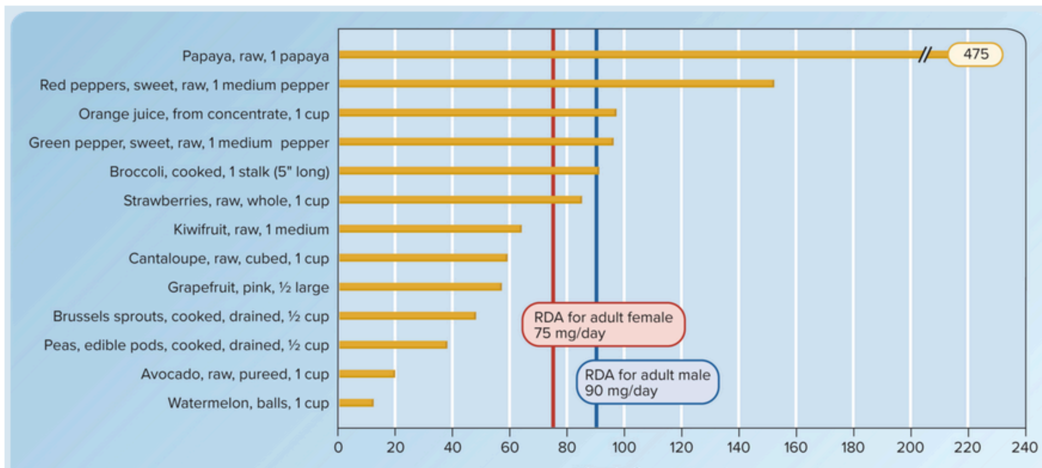
$$\text{DFE} = 100\ \mu\text{g} + (1.7 \times 200\ \mu\text{g})$$
$$\text{DFE} = 100\ \mu\text{g} + 340\ \mu\text{g}$$
$$\text{DFE} = 440\ \mu\text{g}$$

    -
- Dietary adequacy
    - RDA = 400 micrograms, or 600 if pregnant, or 500 if breastfeeding
    - Deficiency
        - Risk increases during periods of rapid growth (pregnancy, infancy, childhood), or with excessive alcohol consumption and certain meds
        - Negative impact on rapid cell division (red blood cells)
        - Bone marrow releases abnormally large RBCs into bloodstream before they mature (megaloblastic anemia) that do not carry normal amounts of oxygen and still have nuclei
        - Neural tube defects (!!)
            - Spina bifida
        - Anencephaly
            - Brain malformation, quickly fatal in most cases
    - Toxicity
        - UL: 1000 micrograms a day (synthetic)
        - Can “mask” a B12 deficiency
        - *May* build up in blood and cause health problems over time

Vitamin C

- Functions
    - Nutrient cofactor that facilitates chemical reactions (collagen synthesis, antioxidant activity, immune function)
    - Synthesis of bile and some neurotransmitters
    - Increases absorption of iron
- Food sources
    - Peppers, citrus, papaya, broccoli, cabbage, berries, potatoes, fortified drinks and cereals

      TABLE 10.10 Vitamin C Content of Selected Foods

    - 

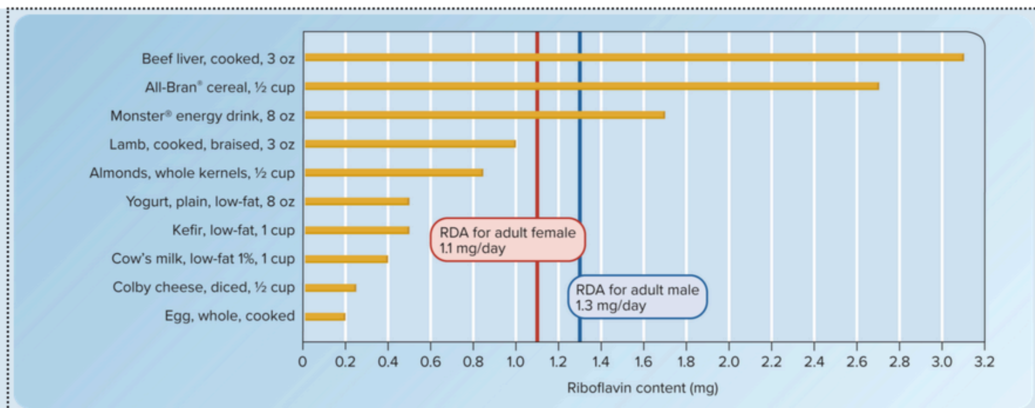
    - Very unstable in heat/oxygen/light/alkaline conditions/presence of iron and copper
- Dietary adequacy
    - RDA = 75 mg for females, 90 mg for males. +35 mg for smokers
    - Deficiency
        - Scurvy
            - Weak connective tissue
            - Unable to maintain existing collagen
            - Pinpoint bleeding under skin (petechiae) around hair follicles
            - Bruises easily
            - Spongy gums that bleed easily
            - Swollen joints
            - Teeth fall out
        - Increased risk of infection
        - At risk
            - Smokers
            - Alcoholics
            - Ppl who don’t eat fruits or veggies
    - Toxicity
        - UL = 2000 mg
        - GI upset

        - Raised risk of kidney stones

Riboflavin

- Functions
    - Component of flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD)
        - Coenzymes with key roles in fatty acid and folate metabolism
        - Small amounts of riboflavin are stored in the heart, liver, and kidneys. Any excess is excreted in urine
- Food sources
    - Cows milk, yogurt, dairy products, enriched cereal, almonds, liver, mushrooms, broccoli, asparagus, spinach, leafy greens

      TABLE 10.3 Riboflavin Content of Selected Foods

    - 

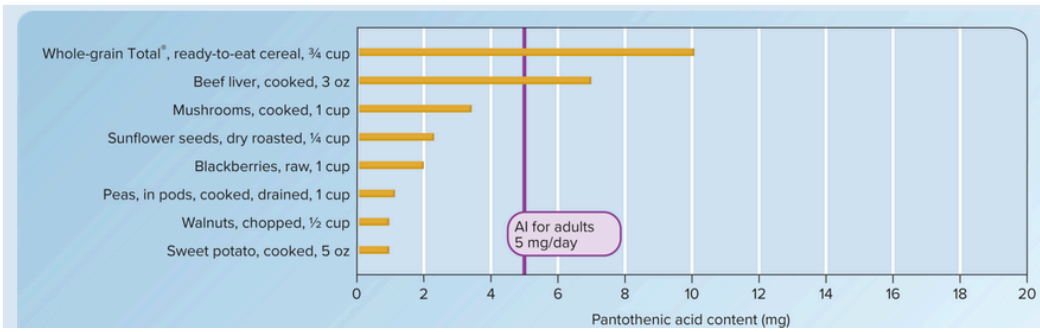
    - Sensitive to light, so products are usually in opaque containers
- Dietary adequacy
    - RDA: 1.1 mg for females, 1.3 mg for males
    - Deficiency
        - Ariboflavinosis (rare)
            - Fatigue
            - Inflammation of mucous membranes in mouth and throat
            - Swollen and sore tongue (lumpy and reddish purple)
            - Chapped lips
            - Scaling and cracking in corners of mouth
            - Skin inflammation
            - Eye disorders
            - Nervous system disorders like headaches and confusion
    - Toxicity
        - No UL, any excess is excreted

Pantothenic acid

- Functions

    - Part of coenzyme that releases energy from macronutrients and is needed for fatty acid synthesis
- Food sources
    - Fortified cereals, liver, sunflower seeds, mushrooms, peas, walnuts, berries, meat, cows milk, many veggies

TABLE 10.5 Pantothenic Acid Content of Selected Foods

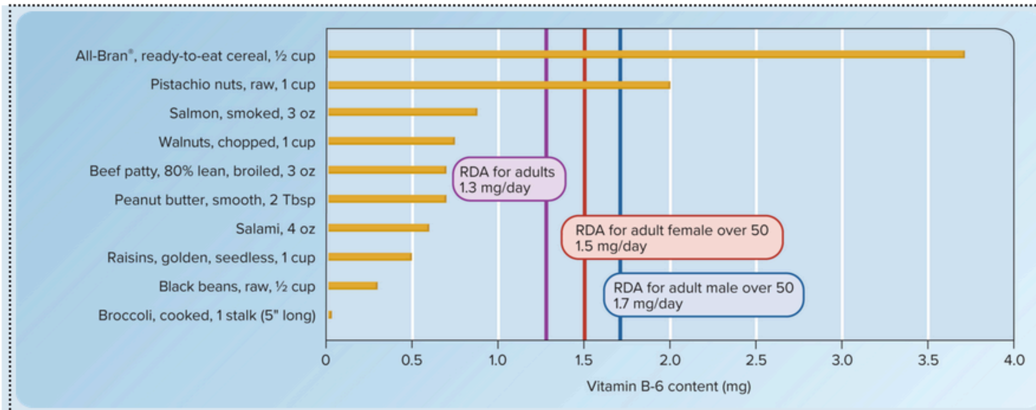


    -
- Dietary adequacy
    - AI = 5 mg a day
    - Deficiency
        - Rare, this stuff is in a whole bunch of foods
        - Headache
        - Fatigue
        - Impaired muscle coordination
        - GI tract disturbances
        - Burning feet syndrome
        - Common in alcoholics
    - Toxicity
        - No UL, no reports of toxicity

Vitamin B6

- Functions
    - Coenzyme, facilitate enzymatic reactions involved in amino acid metabolism (tryptophan → niacin, transmission reactions the form non essential amino acids)
    - Production of heme during red blood cell production
        - Lack can result in anemia
    - Synthesis of neurotransmitters
- Food sources
    - Pistachios, walnuts, salmon, beef, potatoes, bananas, spinach, sweet red peppers, broccoli
    - Lost during grain refinement, not added back in during enrichment, but can be added via fortification
    - Sensitive to excessive heat

TABLE 10.7 Vitamin B-6 Content of Selected Foods

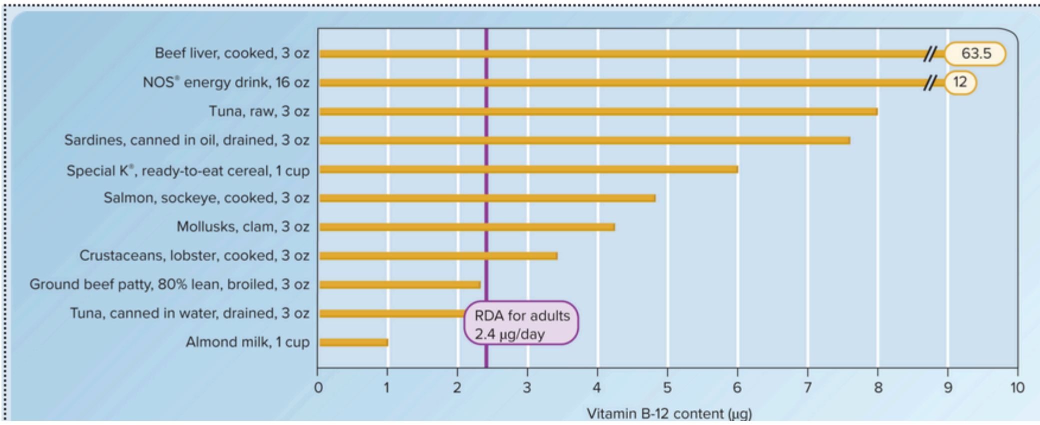


-
- Dietary adequacy
    - RDA: 1.3 - 1.7 mg a day
    - Deficiency
        - Rare in US, but can happen in alcoholics/ppl with genetic conditions that impact B6 metabolism/ppl taking meds for parkinson's disease or tuberculosis
        - Dermatitis
        - Anemia
        - Convulsions
        - Depression
        - Confusion
        - Seizures in people with epilepsy
    - Toxicity
        - UL = 100 mg a day
        - Severe sensory nerve damage (temporary, resolve after they stop taking megadoses)
        - Walking difficulties
        - Numbness of hands and feet

Vitamin B12

- Functions
    - Coenzymes for cell processes, such as methylation in the metabolism of folate
    - Maintaining the myelin sheath
    - Homocysteine metabolism
    - Convert folate to coenzyme forms
- Food sources
    - Bacteria, fungi, algae
    - Meat, cows milk, dairy products, poultry, fish, shellfish, eggs, liver

TABLE 10.9 Vitamin B-12 Content of Selected Foods

- 

- Dietary adequacy
    - RDA: 2.4 micrograms a day
    - Deficiency
        - At risk (do not apply to synthetic forms)
            - Vegans
            - Older adults (not able to release it from the animal proteins as efficiently)
            - Alcoholics
            - Gastric bypass surgery for obesity
            - Gastritis
            - Some meds that lower stomach acid secretion
            - Autoimmune disorders that prevent IF from being made
                - This one is deadly, leads to pernicious anemia. Requires b12 injections monthly or nasal gel containing the vitamin
        - Nerve damage
        - Megaloblastic anemia
        - Muscle weakness
        - Smooth and shiny tongue
        - Confusion
        - Difficulty walking and maintaining balance
        - Numbness and tingling sensations in hands and feet
    - Toxicity
        - No UL

Skim

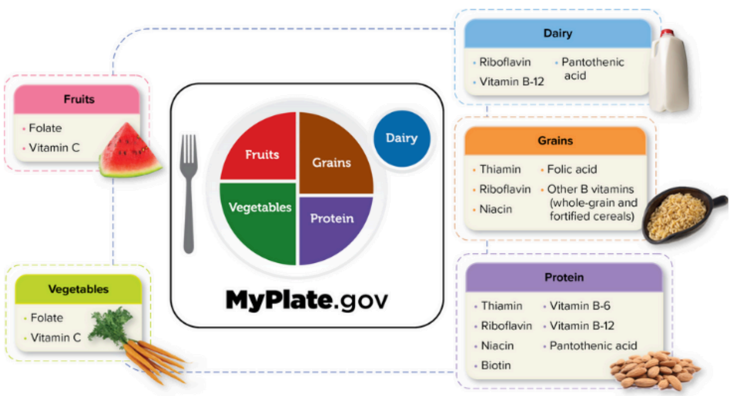
Intro to water soluble vitamins

Dissolve into water, get filtered by kidneys, eliminated through the urine. Need to be consumed regularly since they are not really stored

TABLE 10.1 Summary of Water-Soluble Vitamins

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Thiamin	Part of TPP, coenzyme needed for carbohydrate metabolism and the metabolism of certain amino acids; may help produce neurotransmitters	**1.1–1.2 mg**	Pork, wheat germ, enriched breads and cereals, legumes, nuts	Beriberi and Wernicke-Korsakoff syndrome: weakness, abnormal nervous system functioning	None (Upper Level [UL] not determined)
Riboflavin	Part of FMN and FAD, coenzymes needed for carbohydrate, amino acid, and lipid metabolism	**1.1–1.3 mg**	Cow's milk, yogurt, and other dairy products; enriched breads and cereals	Inflammation of the mouth and tongue, eye disorders	None (UL not determined)
Niacin	Part of NAD and NADP, coenzymes needed for energy metabolism	**14–16 mg**	Enriched breads and cereals, beef, liver, tuna, salmon, poultry, pork, mushrooms	Pellagra • Diarrhea • Dermatitis • Dementia • Death	**Adult UL = 35 mg/day** Flushing of facial skin, itchy skin, nausea and vomiting, liver damage
Pantothenic acid	Part of coenzyme A that is needed for synthesizing fat and that helps release energy from carbohydrates, fats, and protein	5 mg	Beef and chicken liver, sunflower seeds, mushrooms, yogurt, fortified cereals	Rarely occurs	Unknown (UL not determined)
Biotin	Coenzyme needed for synthesizing glucose and fatty acids	30 µg	Eggs, peanuts, salmon, pork, liver, mushrooms, sunflower seeds	Rarely occurs: skin rash, hair loss, convulsions, and other neurological disorders; developmental delays in infants	Unknown (UL not determined)
Vitamin B-6	Part of PLP, coenzyme needed for amino acid metabolism, involved in neurotransmitter synthesis and hemoglobin synthesis	**1.3–1.7 mg**	Meat, fish, poultry, potatoes, bananas, spinach, sweet red peppers, broccoli	Dermatitis, anemia, depression, confusion, and neurological disorders such as convulsions	**Adult UL = 100 mg/day** Nerve destruction

Folate	Part of THFA, coenzyme needed for DNA synthesis and conversion of cysteine to methionine, preventing homocysteine accumulation	**400 µg**	Dark green, leafy vegetables; papayas; asparagus; broccoli; orange juice; enriched breads and cereals (folic acid)	Megaloblastic anemia, diarrhea, neural tube defects in embryos	**Adult UL = 1000 µg/ day** May stimulate cancer cell growth
Vitamin B-12	Part of coenzymes needed for various cellular processes, including folate metabolism; maintenance of myelin sheaths	**2.4 µg**	Animal foods; fortified cereals; fortified almond, soy, or rice milks	Pernicious anemia: megaloblastic anemia and nerve damage resulting in paralysis and death	None (UL not determined)
Ascorbic acid (vitamin C)	Substance needed for connective tissue synthesis and maintenance, antioxidant, synthesis of neurotransmitters and certain hormones, immune system functioning	**75–90 mg** (people who do not smoke cigarettes)	Peppers, citrus fruits, cherries, broccoli, cabbage, berries	Scurvy: poor wound healing, pinpoint hemorrhages, bleeding gums, bruises, depression	**Adult UL = 2000 mg/ day** Diarrhea and GI tract discomfort



Found in food, supplements not really worth the money

Folate: digestion and absorption

No enzymatic activity needed to digest folic acid, but are needed to digest naturally occurring folate (remove glutamates, so it is not bioavailable as folic acid).

3 carbon methyl group is added to folic acid, which is then absorbed by the small intestine. Folate is then activated within cells by the removal of the methyl group (requires B12).

B12: digestion and absorption

- Natural forms bound to animal protein, prevents absorption until released by HCL in stomach gastric juice
- Synthetic forms are more readily absorbed
- Absorbed in the small intestine after binding to intrinsic factor (IF), then separates and travels through bloodstream via transport molecules, to the liver via the hepatic portal vein

- Liver removes b12 from carrier molecules and stores about half of it (50%), enough for generally 5-10 years is stored up